INTERIM FEDERAL SPECIFICATION

MAT, FLOOR, CELLULOSIC PILE YARNS, FLAME RETARDANT

(RENTAL SERVICE)

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D C. 20406, based upon currently available technical information It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1 SCOPE AND CLASSIFICATION

1 1 Scope. This specification covers the requirements for laundered and treated floor mats to be furnished to the Government by commercial services on a rental basis. The mats shall be of the non-skid and non-slip type with cellulosic pile yarn surface and flame retardant.

1.2 Classification

1.2 1 Type and style The floor mats covered by this specification shall be of one type and style

1.2.2 Sizes Unless otherwise specified, the sizes shall be as follows, as specified in the contract or order (6.2. 6 3 and 6 4)

Size Number	Size
1	2 ft 4 5 in by 3 ft
2	2 ft. 6 in. by 3 ft
3	3 ft by 4 ft 9 in
4	3 ft by 5 ft 2 in
5	3 ft by 12 ft.
6	4 ft 9 in by 8 ft
7	3 ft by 16 ft
8	4 ft. 9 in. by 10 ft.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Standard.

Fed Test Method Std No. 191 - Textile Test Methods,

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D C., Atlanta, Chicago Kansas City, Mo , Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard·

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation or bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

D-92-57 - Test for Flash and Fire Points by Cleveland Open Cup
D-418-68 - Woven Pile Floor Coverings
D-1486-57T - Tufted Pile Floor Coverings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3 REQUIREMENTS

3.1 Material.

3.1.1 Pile yarn The pile yarn shall be made of 100 percent cellulosic fibers The pile yarn shall be natural or vat dyed, or a blend of natural and vat dyed fibers Functional equivalents to vat dyes, such as naphthols, will be acceptable.

3.1.2 Backing Backing yarns and backing materials shall be constructed and of proper weight to secure pile yarns in position with strong tuft bind and provide a firm foundation for the mat

3 1 3 Back coating The back coating compound shall be an oil resistant, wash fast, skid and heat-resistant latex compound

3.1.4 Thread The thread used for overedging or serging shall be nylon or cotton Nylon thread shall be 630 denier and cotton shall be of comparable quality

3.1 5 Dust attracting compound The compound for imparting dust attracting properties to the mat pile shall have a paraffinic base with an Underwriters' Fire Hazard Rating between 0-20. The Flash Point (Cleveland Open Cup) shall be no less than 410°F Compound used shall not cause curling of finished mat

3 1 6 Flame retardant compound The flame retardant compound shall contain no volatile solvents and shall not impart any objectionable odor to the mat The compound shall contain no ingredients which (1) are known to be skin sensitizers or irritants and (2) would cause the mats to sour or mildew when stored under normal conditions

3 2 Construction and treatment. The mat shall be of woven or tufted (backstitching may be in either direction) carpet construction and shall have an overall pattern of cut or loop pile face The mat shall be treated by impregnating with a dust control compound and flame retardant compound The mat shall be overedged or serged as specified in 3.2.1 Back coating shall be applied to the back or floor side of the mat to render it skid resistant The pile yarn and tufts shall be firmly secured to the backing materials or the finished mat The finished mat shall

a. Be free from objectionable odor
b. Have no tendency to sour or mildew when stored under normal conditions.
c. Show no tendency to heat spontaneously under normal storage conditions
d. Be free of any ingredients known to be skin sensitizers or irritants.

3.2.1 Overedging or serging All cut edges shall be overedged or serged with thread as specified in 3.1.4. The end of the overedging or serging shall be bonded or otherwise secured to avoid unraveling of the stitching. The edges of the finished mats shall lay firmly and flat on the floor surface and shall be resistant to curl when tested as specified in 4.3. Nosings, or their equivalent, may be used to meet this requirement (i.e lay flat, resistant to curl)

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3.3 Physical requirements. The physical requirements for new untreated mats shall be as specified in table I and for the finished treated mat shall be as specified in table II.

TABLE I. Physical requirements of new untreated mat

Characteristic	Requirement
Weight, pile yarn, ozs sq. yd., minimum 1/	21.5
Weight, overall, ozs sq. yd., minimum 1/	35.0
Back coating, ozs sq. yd., minimum	10.5

1/ Exclusive of back coating

TABLE II. Physical requirements of finished treated mat

Characteristic	Requirement
Weight, dust attracting compound, percent of total weight of pile	
Minimum	6.0
Maximum	14.0
Flame resistance, longest diameter of charred area, inches, maximum	3.0
Pile height or tuft length, inches, minimum	0.3

3.4 Size and tolerances. The size or size number (1.2.2) of the mats shall be as specified in the contract or order (6.2). Tolerances for both the width and length of mat shall be plus or minus 5 percent.

3.5 Color. Unless otherwise specified in the contract or order (6.2), the color of the pile surface of the mat shall be at the option of the contractor.

3.6 Identification marking. Each mat when new shall be marked or labeled clearly to identify the supplier by name or trademark of such known character that the source of supply can be readily determined.

3.7 Workmanship. The finished mat shall be clean and free from tears, holes, ragged edges, loose ends or ravelings. There shall be no distortions or bulges of the mat when laid on the floor or other imperfections which detract from their appearance or may impair their serviceability. When the mats are laid on the floor, the mat edges shall lay flat on the floor surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used meet the requirements of this specification. Testing shall be performed on the components and materials listed in table III. The sample unit shall be of sufficient size to conduct the required tests. The sample size shall be as specified in table IV. The lot shall be unacceptable if one or more sample units fail to meet any specified requirement. The lot size shall be expressed in terms shown in table III.

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TABLE III Component and material testing 1/

Component and lot size expressed in terms of	Characteristic	Requirement reference	Test method
Pile yarn (lbs)	Material identification	3 1.1	Comm Std
Back coating compound (pt)	Material identification	3.1 3	Comm Std
Thread (lbs)	Material identification	3 1 4	Comm. Std.
	Denier		4021 2/
Dust attracting compound (pt)	Fire hazard rating	3 1 5	1/
	Flash point	3 1 5	ASTM D-92-57

1/ May be accepted based on contractor's certification of compliance
2/ Fed Test Method Std No 191.

TABLE IV. Sample size

Lot size (units)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

4 2.2 Intermediate inspection Testing shall be performed on new untreated mats for the characteristics specified in table V The sample unit shall be of sufficient size to conduct the required tests The sample size shall be as specified in table VIII. The lot size shall be acceptable if one or more sample units fail to meet any specified requirement The lot size shall be expressed in units of mats each Because of the nature of mat rental contracts where repeated treatments may be required on the mats during their service life, new untreated mats may not be available at the start of a contract period for testing. Acceptance may be based on certification of compliance by the contractor that the untreated mats when new complied with requirements and characteristics in table V

TABLE V Testing of new untreated mat 1/

Characteristic	Requirement reference	Test method
Weight pile yarn	Table I	4 4 4
Weight, overall	Table I	5040 or 5041 2/
Weight back coating 3/	Table I	4.4 5

1/ May be accepted based on contractor's certification of compliance
Fed Test Method Std No 191 When method 5041 is used, one specimen of at least 10 square inches shall be used
3/ The test results for this characteristic shall be reported as the lot average

4.2 3 Examination of the end item The finished mat shall be examined for the defects listed in table VI The inspection level shall be level I and the acceptable quality level shall be 2.5 defects per hundred units The sample unit for this examination shall be one mat The lot size shall be the number of mats submitted for delivery at one time The Government reserves the right to perform 100 percent inspection of mats furnished under this specification.

TABLE VI Examination for defects

Examine	Defects
Material and workmanship	Any hole or tear Any specified component missing Not constructed as specified. Back coating missing Edges not bound as specified. Pile yarn not firmly secured. Tufts easily pulled from pile surface Ragged edge. Loose ends or raveling. Distortions or bulges which would prevent the mat from laying flat on a floor (including mat edges). No identification marking (when mat is new,

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TABLE VI. Examination for defects (Cont'd)

Examine	Defects
Finish and appearance	Spots or stains seriously affecting appearance. Discolored areas seriously affecting appearance (variations in color due to crushing of pile shall not be considered a defect) Glazed, resinified or hard backs resulting in loss of skid resistance. Objectionable odor.

4.2.4 Dimensional (length and width) examination. The mats shall be examined for defects in dimensions. The inspection level shall be S-3 of MIL-STD-105. Sample unit and lot size shall be as specified in 4.2.3. Any length or width measurement not in conformance with dimensions, including tolerances, specified herein shall be classified as a defect. The acceptable quality level shall be 1.5 defects per hundred units.

4.3 Testing of the end item. Tests shall be performed for the characteristics specified in table VII. The methods of testing specified in table VII shall be followed. The lot size shall be expressed in units of one mat. The sample size shall be unacceptable if one or more sample units fail to meet any specified requirement.

TABLE VII. End item testing

Characteristic	Requirement reference	Test method
Pile height	Table II	4.4.1
Weight, dust attracting compound 1/	Table II	4.4.6
Weight, flame retardant compound 1/	Table II	Comm. Std.
Flame resistance	Table II	4.4.2
Resistance to curl	3.2.1	4.4.3

1/ Acceptance of this characteristic may be based on a contractor's certification of compliance, accompanied by actual test, inspection or other verifiable quality data.

4.3.1 Lot and sample size for end item testing. The lot size shall be the number of mats submitted for delivery at one time. The sample size for testing of all characteristics, except flame resistance, as listed in table VII, shall be as specified in table VIII. For flame resistance tests, the sample size shall be one (1) for 50 or less mats in the lot, two (2) for 51 to 500 mats in the lot and two (2) for every multiple of 500 mats or less in lots of 501 and over.

TABLE VIII. Sample size

Lot size	Sample size
250 or less	1
251 to 1,000	2
1,001 to 5,000	3
5,001 to 10,000	4
10,001 and over	5

4.4 Tests

4.4.1 Pile height. Pile height shall be determined by folding the mat on itself, with the pile out forming a crease where the measurement is to be taken. Smooth and straighten the pile by hand and measure with a stiff metal ruler. The average of three measurements shall be considered the mat pile height.

4.4.2 Flame resistance. Tests shall be conducted under controlled conditions of 70° (+ 2°) F. and 65 percent (+ 2 percent) relative humidity. Two 6 inch by 6 inch specimens which have been treated at least 24 hours prior to testing shall be tested. The two specimens shall comprise

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mple unit. When more than one sample unit is tested no more than 2 specimens shall be taken any one mat. The equipment required shall be

(1) Methenamine Timed Burning Tablets, Eli Lilly No 1588 Reagent Tablet.
(2) Ruler, graduated in inches and tenths of inches
(3) Matches

procedure shall be as follows

specimen shall be placed flat, with the pile up, in a draft free test area A methenamine et shall be placed on top of the pile and centered on the sample The tablet shall be ignited fully with a match which shall be removed as soon as ignition has been achieved The tablet mat, if it ignites, shall be allowed to burn until self extinguished. The longest diameter of charred area shall be measured to the nearest tenth of an inch on each specimen and the results aged In the event of dispute, additional specimens shall be tested after conditioning for ours at a temperature of 70° (± 2°) F. and relative humidity of 65 percent (± 2 percent).

4 3 Resistance to curl A sample of the finished mat shall be laid on the floor There l be no curling condition, in that the mats shall lay flat on the floor at all points around edges In addition, the edges of the finished mat shall be bent or folded by hand ten times simulate kicking or pushing up of the edge as might occur in normal traffic). The mat should rt to natural position, i e flat on the floor at all points.

4 4 Weight of pile yarn The pile yarn shall be separated from other yarns on a specimen aining 16 square inches To express weight of pile yarn for mat (36 by 36), the weight of pile shall be calculated as follows Weight of pile yarn (ounces) in 16 square inch sample) = weight of pile yarn in ounces per square yard of mat Pile yarn shall be separated from r yarns prior to back coating

4 5 Weight of back coating The weight of the mat after back coating shall be determined in rdance with 5041 of Fed Test Method Std No. 191. Specimen shall have an area of at least 16 re inches. The results of this test shall be reported as the lot average The individual sample results obtained fro the total weight characteristics, exclusive of back coating, shall be average this result subtracted from the lo. average weight after back coating. The difference obtained be the lot average weight of the back coating and shall be reported to the nearest ounce

4 6 Dust attracting compound Clip from the mat approximately 10 grams of pile. Allow clippings to condition in an atmosphere of 65% relative humidity and 70° F overnight After itioning, weigh the clippings to the nearest tenth milligram Extract the clippings in a able seated in a Soxhlet Extractor, using toluene, for one hour. Separate the toluene from extracted compound and weigh the extract to the nearest tenth of a milligram Subtract the extract weight from the weight of the (B) yarn clippings and divide this result into the ght of the extract to determine the weight of the dust attracting compound as percent of total ght of the pile, as follows

$$\% \text{ Treatment} = \frac{A}{B - A} \times 100$$

5 PREPARATION FOR DELIVERY

5 1 Packaging, packing and marking Unless otherwise specified in the contractor or order 2), packaging, packing and marking shall be in accordance with standard commercial practice

6 NOTES

6.1 Intended use The mats covered by this specification, to be furnished to the Government commercial services or a rental basis, are intended for use in entrance ways, corridors, lobbies similar areas They can be used to reduce tracking of dirt, water, snow, grit and other taminates into the interior of buildings, to help prevent slips and falls to protect floors to reduce noise in the applicable areas.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and curement documents should specify the following

(a) Title, number and date of this specification
(b) Size required (1 2 2, 3 4 and 6.4).
(c) Color, if other than as specified (3.5).
(d) Packaging, packing and marking requirements, if other than as specified (5.1)

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6.3 Commercial size numbers Some segments of the industry use a commercial numbering system for the sizes referenced in 1.2.2. The equivalents are as follows

Size Number	Commercial Size Number
1	7
2	7a
3	15
4	15a
5	36a
6	40
7	46a
8	50

6.4 Size required The sizes listed in 1.2.2 are most frequently ordered. Sizes other than those listed may be ordered by the purchasing activity as required.

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